Department of Information Science and Engineering

List of DVT Lab Programs

Course Title: Data Visualization Tools	**Course Code: IS1553-1**

Course Outcomes: At the end of the course student will be able to
1. Implement different data visualization features
2. Create charts and dashboards using different tool

Course Outcomes Mapping with Program Outcomes & PSO														
Program Outcomes →	1	2	3	4	5	6	7	8	9	10	11	12	PSO	
Course Outcomes													1	2
IS1553-1.1	1	2	2		2			1	1		1		3	2
IS1553-1.2	1	2	2		2			1	1		1		3	2
1: Low 2: Medium 3: High														

Sl. No.	**Experiment**	**CO**
1	Create a data sheet of 10 employees with five columns Sl. No., Name, Date of Birth, Gender and Total salary. a. Format the different column values for consistency with respect to currency and date formats. b. Add a column age (Calculate the age using DOB and Current date) c. Calculate the income tax based on the salary. Use the following table to calculate IT. Range: Income tax 0-4 lakh rupees: Nil 4-8 lakh rupees: 5% above 4,00,000 8-12 lakh rupees: 20,000+10% above 8,00,000 12-16 lakh rupees: 60,000+15% above 12,00,000 16-20 lakh rupees: 1,20,000+20% above 16,00,000 20- 24 lakh rupees: 2,00,000+25% above 20,00,000 Above 24 lakh rupees: 3,00,000+30% above 24,00,000 d. Count the number of female employees paying income tax e. Count the number of male employees exempted from paying tax f. Add a column net salary which is calculated after deducting tax amount from Total salary	**1**
2	Create a data sheet pertaining to sales data of a company for 10 years starting from 2006 till 2020. a. Calculate Mean, Median and Mode of sales b. Calculate standard deviation and variance of sales c. Calculate coefficient of correlation between year and sales. Interpret the result d. Fit a regression line for sales against year. Predict the sales for the year 2025.	**1**

	e. Calculate all descriptive statistics for the data sheet	
3	Create 5 data sheets as described below. Draw the line chart, scatter plot, bar chart and Pie Chart relevant to the data in each sheet. a. Semester-wise SGPA of a student (8 semesters). b. Semester-wise pass percentage of a section (8 semesters) c. Yearly sales of a company from 2015 to 2024 d. Age of 100 persons (use random function to get the age of a person). e. Monthly expenditure details (in Rs.) of a person. Consider the items Consumables, EMI, Travel, Utility bills (water, electricity, gas etc.), Investment or saving Schemes (Insurance, Stock, Mutual Fund etc.), Entertainment, and other.	**2**
4	Consider the data set Position_Salaries.csv. Using Python Do the following tasks a. Display first and last few rows of the data b. Display selected columns Position and salary c. Display the rows with salary >90,000 d. Create a column TDS with TDS=0.10*Salary e. Sort the dataset based on Position f. Group the dataset by level and calculate the mean salary of each group g. Plot the graph Level vs Salary	**1, 2**
5	Consider the data set Social_Network_Ads.csv. Using Python Do the following tasks a. Calculate the average, sample variance, sample median and quartiles of the variable "Age" b. Calculate the average, sample variance, sample median and quartiles of the variable "salary" c. Plot the histogram of age and frequency d. Find the outliers with threshold value 3. (outliers are those data values for which the absolute value of data – mean is greater than the product of threshold value and standard deviation). Replace the outliers by mean. Plot the histogram after outliers are replaced.	**1, 2**
6	Create dashboards in Tableau Public Edition using a COVID-19 (compact) dataset: a. Global COVID-19 Tracker Dashboard: - Create a dashboard to track global COVID-19 cases, deaths, and recoveries i. Map to show geographic distribution of cases ii. Bar chart to show top 10 countries with most cases iii. Line chart to show daily new cases and deaths Add filters for date range and country/region b. COVID-19 Vaccination Progress Dashboard i. Map to show geographic distribution of vaccination rates ii. Bar chart to show top 10 countries with highest vaccination rates iii. Line chart to show daily new vaccinations Add filters for date range and country/region c. COVID-19 Hospitalization and ICU Dashboard i. Map to show geographic distribution of hospitalization rates ii. Bar chart to show top 10 countries with highest hospitalization rates iii. Line chart to show daily new hospitalizations and ICU admissions d. COVID-19 Testing and Contact Tracing Dashboard i. Bar chart to show top 10 countries with highest testing rates	**2**

	ii. Line chart to show daily new tests and contact tracing efforts iii. Map to show geographic distribution of testing and contact tracing efforts	
7	Create dashboards in Tableau Public Edition using the sample - Superstore dataset: a. Sales Performance Dashboard i. Total Sales by Region: A map showing the total sales by region. ii. Sales by Product Category: A bar chart showing the sales by product category. iii. Top 5 Products by Sales: A table showing the top 5 products by sales. iv. Monthly Sales Trend: A line chart showing the monthly sales trend. b. Product Analysis Dashboard i. Product Distribution by Category: A Treemap showing the product distribution by category. ii. Product Distribution by Sub-Category: A Treemap showing the product distribution by sub-category. iii. Top 5 Products by Profit: A table showing the top 5 products by profit. iv. Product Sales by Quarter: A bar chart showing the product sales by quarter. c. Time-Based Analysis Dashboard i. Monthly Sales Trend: A line chart showing the monthly sales trend. ii. Quarterly Sales Trend: A line chart showing the quarterly sales trend. iii. Yearly Sales Trend: A line chart showing the yearly sales trend. iv. Top 5 Quarters by Sales: A table showing the top 5 quarters by sales. d. Profitability Analysis Dashboard i. Profit by Product Category: A bar chart showing the profit by product category. ii. Profit by Region: A map showing the profit by region. iii. Top 5 Products by Profit: A table showing the top 5 products by profit. iv. Gross Margin by Quarter: A line chart showing the gross margin by quarter.	**2**